

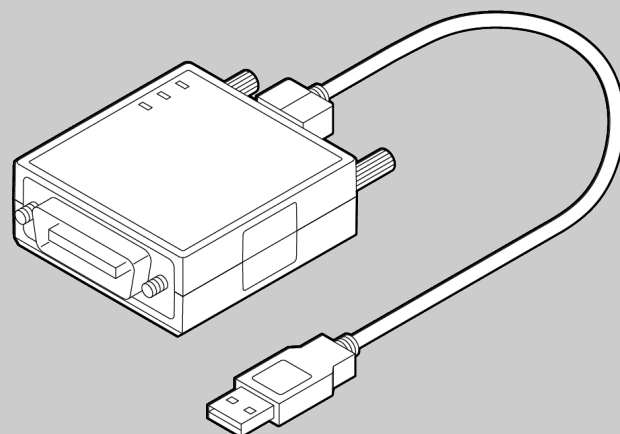
# Reference Manual

High Speed IEEE-488.2 I/F Micro Converter for USB2.0

## GPIB-FL2-USB

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# Introduction

This section provides necessary information of the product such as the outline, bundled items and manuals before actual use.

# 1.Related Manuals

The manuals related to the product are listed below.  
Read them as necessary along with this document.

## ◆ Must Read the Followings.

| Name                             | Purpose                                   | Contents                                                                                                                                      | How to get                                                                                                                 |
|----------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Please read the following        | Must read this after opening the package. | This introduces related materials that are made available on the CONTEC website, such as those for the included items, manuals, and software. | Included in the package (Printed matter)                                                                                   |
| Reference Manual (This Document) | Read this when operating the product.     | This describes the hardware aspects such as functions and settings.                                                                           |  Download from the CONTEC website (PDF) |

## ◆ Download Manuals

Download the manuals accordingly from the following URL.

**Download**

<https://www.contec.com/download/>

## 2.About the Product

This product is a micro converter that converts USB port of a PC to a GPIB communication port compatible to IEEE-488 standard.

With the USB cable (1.8 m) included in this package, it can be connected to GPIB interface of any equipment without using GPIB cables. Furthermore, this product is powered by the bus so that simple and compact GPIB communication system can be established.

Windows/LabVIEW device driver is supported with this product.

## 3.Features

### ■ Capable to communicate with GPIB communication which is compatible to IEEE-488.1/488.2 standards at 1.5 M byte/sec maximum

Capable to communicate with any equipment which is compatible to IEEE-488.1/488.2 standards with transfer rate at 1.5 Mbyte/sec maximum. Capable to set this product to whether a master (controller) or a slave.

### ■ Compatible to USB2.0/USB1.1 and bus power-driven eliminating the need for external power.

Compatible to USB2.0/USB1.1 and capable to achieve high speed transfer at HighSpeed (480 Mbps).

The product is driven by bus power from USB, which eliminates the need for external power. In addition, with the USB cable (1.8 m) included in this package, it can be connected to any equipment with GPIB interface without any GPIB cables.

### ■ Employs a buffer memory, 2 Kbytes for transmission and 2 Kbytes for reception.

Employs a buffer memory, 2 Kbytes dedicated to transmission and 2 Kbytes dedicated to reception, in order to reduce the load to the CPU when transmitting/receiving data.

### ■ Windows/LabVIEW compatible support software is offered.

The support software offered on the CONTEC website makes it possible to create applications of Windows/LabVIEW. In addition, supplies a diagnostic program to confirm hardware operation and to perform a basic communication test with connected equipment.

### ■ Employs a high speed GPIB controller developed by CONTEC and provides steady-supply

This product employs CONTEC's self-developed high speed GPIB controller ( $\mu$ PD7210 register-compatible), which provides users steady-supply with peace of mind.

### ■ Built-in SPAS event function (when slaving)

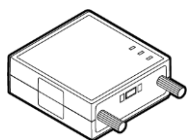
In addition to the functions of the earlier GPIB controller ( $\mu$ PD7210), the product also supports the SPAS event generated when a serial poll occurs. This gives users a high level of flexibility in constructing the system.

## 4. Product Configuration List

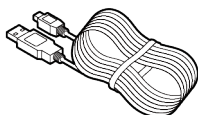
The product consists of the items listed below.

Check, with the following list, that your package is complete.

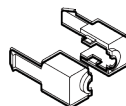
If you discover damaged or missing items, contact your retailer.



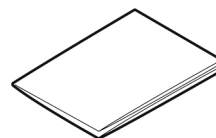
Product...1



USB Cable (1.8m)...1



USB Cable Attachment on  
the main unit's side...1



Please read the  
following...1

## 5.Support Software

You can use CONTEC support software according to your purpose and development environment. For more details on the supported OS, applicable languages, or to download the latest version of software, visit the CONTEC's Web site.

| Name                                                             | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                         | How to get                       |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Driver software<br>API-GPIB(WDM) for USB                         | It is the driver software for Windows with new function specification and supplies command in the form of standard Win32 API function (DLL).<br>Ideal for a user who intend to construct a new system since setting up multiple GPIB communication converters is easy, for example.<br>In addition, sample programs such as Visual Basic and Visual C++ are included, and you can verify the operation of hardware by using Diagnostic programs. | Download from the CONTEC website |
| Driver software<br>API-GPIB(98/PC) *1                            | It is the driver software, and which supplies command in the form of standard Win32 API function (DLL).<br>In addition, sample programs such as Visual Basic and Visual C++ are included, and you can verify the operation of hardware by using Diagnostic programs.                                                                                                                                                                             | Download from the CONTEC website |
| GPIB communication driver<br>supporting LabVIEW<br>API-GPLV(W32) | API-GPLV(W32) is a driver created according to the National Instruments Corporation's GPIB function style.<br>The driver is software to control the CONTEC GPIB board using a LabVIEW-based GPIB system or existing application program.                                                                                                                                                                                                         | Download from the CONTEC website |

\*1 When replacing our GPIB board (F series) with this product.

Download the files accordingly from the following URL.

Download

<https://www.contec.com/download/>

## 6.Optional Products

Optional product items are as follows:

Use these items with the main unit as necessary.

| Product Name           | Model type | Description |
|------------------------|------------|-------------|
| GPIB cable             | PCN-T02    | 2m          |
|                        | PCN-T04    | 4m          |
| GPIB connector adapter | CN-GP/C    | *1          |

\*1 Effective if this product interferes with the main unit of the target device when plugging this product into the device.

Visit the CONTEC website for the latest optional products.

**Website**

<https://www.contec.com/>

# Safety Precautions

Understand the following definitions and precautions to use the product safely.

Never fail to read them before using the product.

# 1.Safety Information

This document provides safety information using the following symbols to prevent accidents resulting in injury or death and the destruction of equipment and resources.

Understand the meanings of these labels to operate the equipment safely.

|                                                                                                  |                                                                                                                                 |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|  <b>DANGER</b>  | Signal word used to indicate an imminently hazardous situation which, if not avoided, will result in death or serious injury.   |
|  <b>WARNING</b> | Signal word used to indicate a potentially hazardous situation which, if not avoided, could result in death or serious injury.  |
|  <b>CAUTION</b> | Signal word used to indicate a potentially hazardous situation which, if not avoided, could result in minor or moderate injury. |

## 2. Handling Precautions

### **DANGER**

- Do not use the product in locations exposed to a flammable or corrosive gas. It may cause explosion, fire, electrical shock, or malfunction.

### **CAUTION**

- Do not strike or bend this product.  
Otherwise, this product may malfunction, overheat, cause a failure or breakage.
- Make sure that your PC can supply ample power to all this product installed.  
Insufficiently energized products could malfunction, overheat, or cause a failure.
- It may cause a trouble in recognizing and operating the device according to the kind of USB hub.
- As this product contains precision electronic components, do not use or store it in the environments subject to shock or vibration. Otherwise the product may malfunction, overheat, cause a failure or breakage.
- Do not use or store the product in a hot or cold place where the temperature exceeds the specified range, or in a place that is subject to severe temperature changes.  
Otherwise the product may malfunction, overheat, cause a failure or breakage
- Do not use or store the product in a place subject to direct sunlight or near a heating device, such as a stove. Otherwise the product may malfunction, overheat, cause a failure or breakage.
- Do not use or store the product near equipment generating a strong magnetic field or radio waves. Otherwise the product may malfunction, overheat, cause a failure or breakage.
- Do not touch this product's terminals (USB connector, Interface connector) with your hand.  
Otherwise, the product may malfunction, overheat, or cause a failure.  
If the terminals are touched by someone's hands, clean the terminals with industrial alcohol.
- Do not touch the interface connector when the power is on.  
Otherwise the product may malfunction, overheat, cause a failure due to static electricity.
- Do not plug or unplug the internet connector when the power is on. This may cause a failure.
- Do not operate switches when the power is on. This may cause a failure.
- When connecting multiple units, set one unit at a time. Complete the setup before starting to connect the next unit.
- The specifications of this product are subject to change without notice for enhancement and quality improvement.  
Even when using the product continuously, be sure to read the manual and understand the contents.
- Do not modify or disassemble the product. CONTEC will bear no responsibility for any problems, etc., resulting from modifying or disassembling the product.

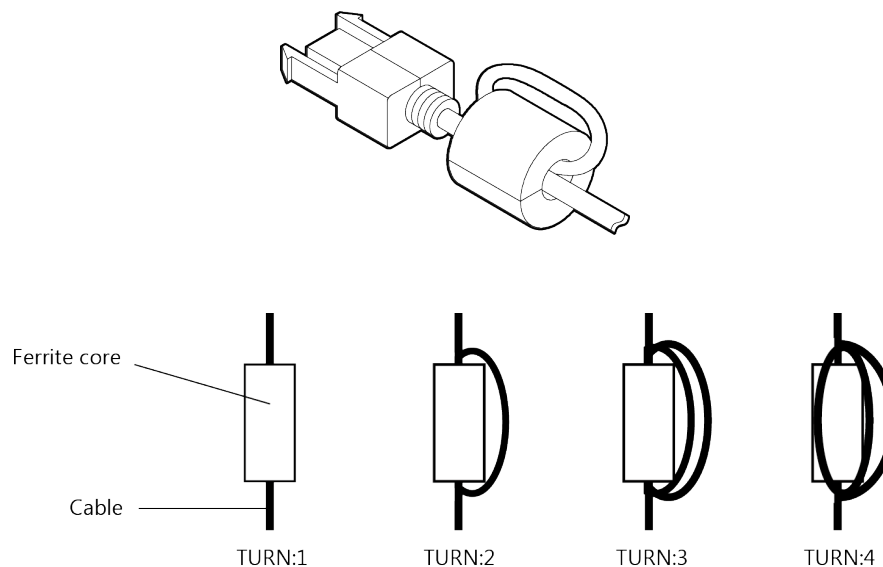
- The product must always be accompanied by the Setup Guide when transferred or assigned.
- Contact your retailer for product repair or replacement.
- When disposing of the product, follow the disposal procedures stipulated under the relevant laws and municipal ordinances.
- Regardless of the foregoing statements, CONTEC is not liable for any damages whatsoever (Including damages for loss of business profits) arising out of the use or inability to use this CONTEC product or the information contained herein.

## 1. Regarding “CE EMC Directive Class A Notice”

The ferrite core must be installed in the connecting cable so that this product may suit the above-mentioned standard. (Equivalent product can also be used)

| Name     | Maker               | Turn | Quantity | Installation Site            |
|----------|---------------------|------|----------|------------------------------|
| RFC-13MA | KITAKAWA INDUSTRIES | 4    | 1        | on USB cable at product side |
| RFC-20   | KITAKAWA INDUSTRIES | 2    | 1        | on GPIB cable at pc side     |

Image diagram



## 2. FCC PART15 Subpart B Class A

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### NOTE

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

### FCC WARNING

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

## 3. EN55032Class A Notice

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Warning:

Operation of this equipment in a residential environment could cause radio interference.

## 3.Environment

Use this product in the following environment. If used in an unauthorized environment, this product may overheat, malfunction, or cause a failure.

### Operating temperature

0 - 50°C

### Humidity

10 - 90%RH (No condensation)

### Corrosive gases

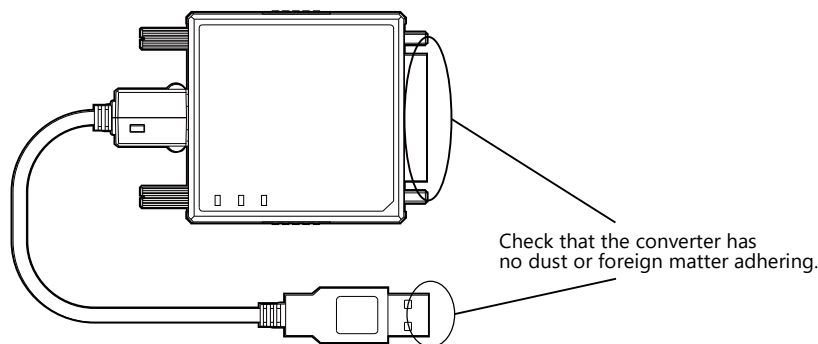
None

### Floating dust particles

Not to be excessive

## 4.Inspection

Inspect the product periodically as follows to use it safely.



## 5.Storage

When storing this product, keep it in its original packing form.

- Put this product in the storage bag.
- Wrap it in the packing material, and then put it in the box.
- Store the package at room temperature at a place free from direct sunlight, moisture, shock, vibration, magnetism, and static electricity.

## 6. Disposal

When disposing of the product, follow the disposal procedures stipulated under the relevant laws and municipal ordinances.

# Setup

This section explains how to set up this product.

# 1.What is Setup?

Setup means a series of steps to take before the product can be used.

Different steps are required for device driver and hardware.

The setup procedure will depend on your combination of OS and device driver. For details, refer to the help for the device driver you will use. This section describes the procedure to start the application program development using the Windows version of the device driver API-GPIB(WDM) for USB.

## 1. Device driver Installation

---

Taking the following steps sets up the device driver and hardware. You can use the diagnosis program later on to check whether the device driver and hardware function properly.

**Step 1 Device driver Installation (page20)**

**Step 2 Hardware Setting (page21)**

**Step 3 Hardware Installation (page25)**

**Step 4 Device driver Initialization (page27)**

**Step 5 Operation Check (page28)**

If Setup fails to be performed properly, see the “**Setup Troubleshooting (page32)**” section at the end of this chapter.

## 2. Device driver Installation

This manual describes how to install the Windows device driver.

Before connecting this product to a PC, install the device driver.

Download the "Device driver API-GPIB(WDM) for USB " from the CONTEC website.

### Download

<https://www.contec.com/download/>

For the device driver installation procedure, refer to the help included in the development environment package you downloaded from the CONTEC website or "**Installing Device Driver**" in the online help made available on the CONTEC website.

### Online Help [Installing Device Driver]

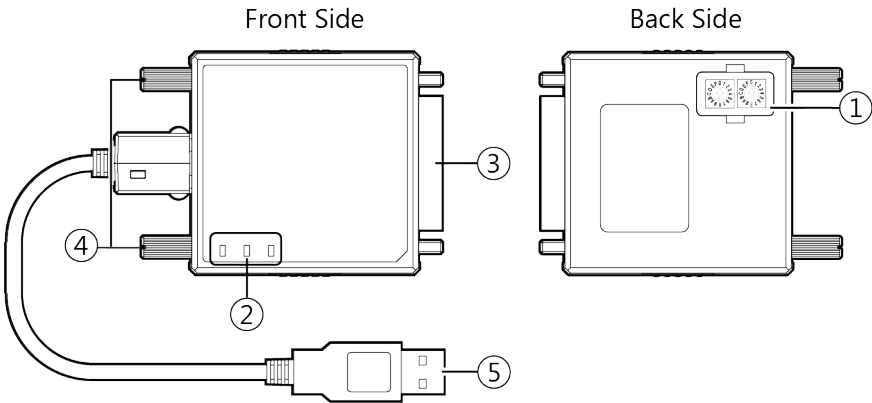
<https://help.contec.com/link/drv/wdm/install/en/>

# 3. Hardware Setting

This section describes how to set up the product and how to connect it to a PC.

## 1. Nomenclature of Product Components

Component names of the product are shown in the figure below.

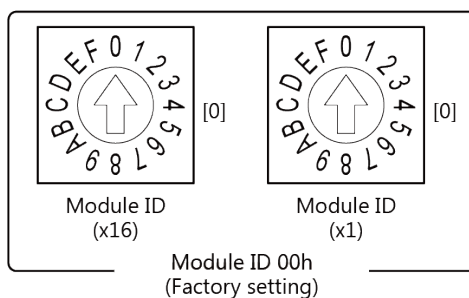


| No. | Name                | Description                                                |
|-----|---------------------|------------------------------------------------------------|
| 1   | Setting Switches    | These are switches to distinguish modules.                 |
| 2   | LED Indicator       | This indicates the state of the product.                   |
| 3   | Interface Connector | This is a 24-pin GPIB connector (Male)                     |
| 4   | Connecting Screws   | These are screws to connect GPIB cable or a remote device. |
| 5   | USB Connector       | This is a USB connector for type-A.                        |

## 2. Setting Switches

With the two setting switches, the host computer distinguishes and keeps track of the devices of same model by assigning Module IDs to them. Factory setting "00" can be used when only one device per model is connected to one computer. Each device should be assigned a unique Module ID in the range of 00 - 7Fh when several devices with the same model are being connected.

"x16" and "x1" represent high bits and low bits of Module ID respectively.



| Module ID | Description                                    |
|-----------|------------------------------------------------|
| 00 - 7Fh  | It is a setting range for module IDs.          |
| FFh       | This is used only when upgrading the firmware. |

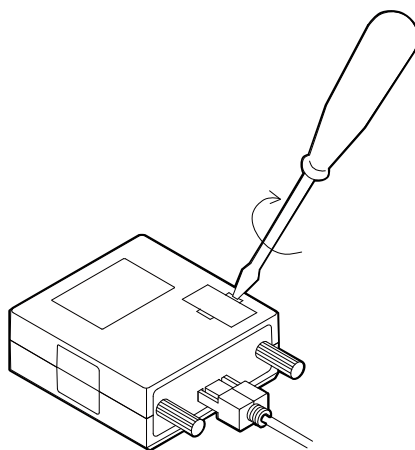
### CAUTION

When setting with Module ID(x1) = F, Module ID(x16) = F, Module ID is only used for upgrading the firmware. This setting cannot be used to distinguish or keep tracks of the devices.

### How to open the cover

Use a flathead screwdriver or similar to open the cover on the underside of the product.

This exposes the setting switches to set the Module ID. As shown in Figure below, an easy way to open the cover is to insert the tip of the screwdriver into the slot and then rotate the screwdriver.



### 3. LED Indicator

The product indicates its operation state by means of light emitting diodes.

The LED indicates as listed below.

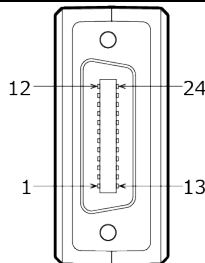


List of Status LED Functions

| LED  | Function                    | Color  | Display |  | Description                          |
|------|-----------------------------|--------|---------|--|--------------------------------------|
| LA   | Status of address           | Yellow | ON      |  | Listener mode                        |
|      |                             |        | OFF     |  | Not listener mode                    |
| SRQ  | Output status of SRQ signal | Yellow | ON      |  | SRQ output (Low)                     |
|      |                             |        | OFF     |  | No SRQ output (High)                 |
| LINK | USB communication status    | Green  | ON      |  | Communication is established.        |
|      |                             |        | OFF     |  | Communication is not established.    |
|      | PC connection status        | Green  | ON      |  | PC communication is established.     |
|      |                             |        | OFF     |  | PC communication is not established. |

### 4. Interface Connector

Use the 24-pin connector attached to this product to connect to a remote device.



Compatible connector (cable): GPIB cable (IEEE-488 standard follow)

| Pin number | Signal name               | Pin number | Signal name         |
|------------|---------------------------|------------|---------------------|
| 12         | GND (Shield)              | 24         | GND (Logic)         |
| 11         | ATN (Attention)           | 23         | GND (Grand)         |
| 10         | SRQ (Service Request)     | 22         | GND (Grand)         |
| 9          | IFC (Interface Clear)     | 21         | GND (Grand)         |
| 8          | NDAC (Not Data Accepted)  | 20         | GND (Grand)         |
| 7          | NRFD (Not Ready for Data) | 19         | GND (Grand)         |
| 6          | DAV (Data Valid)          | 18         | GND (Grand)         |
| 5          | EOI (End Of Identify)     | 17         | REN (Remote Enable) |
| 4          | DIO4                      | 16         | DIO8                |
| 3          | DIO3                      | 15         | DIO7                |
| 2          | DIO2                      | 14         | DIO6                |
| 1          | DIO1                      | 13         | DIO5                |

## 5. Connecting Screws

---

These are screws to connect GPIB cable or a remote device.

## 6. USB Connector

---

This is a USB connector for type-A. Connect this to the USB port on the PC side.

## 4. Hardware Installation

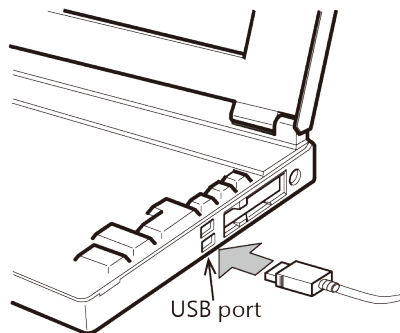
Under Windows, the peripherals need to be recognized by the OS. This is called hardware installation.

**When using multiple converters, install one at a time. Complete the setup of the converter before starting to install the next one.**

### 1. Connecting the Product

- 1 Turn on the power to the PC before connecting this product.
- 2 When the PC has been up and running, connect the product to a USB port in the PC.

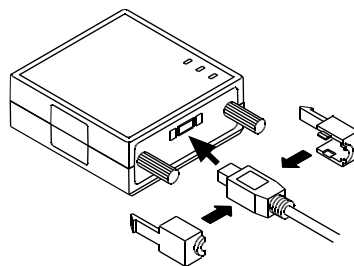
**Connect the product to a USB port in the PC.**



#### ⚠ CAUTION

It may cause a trouble in recognizing and operating the device according to the kind of USB hub.

#### Attaching a USB Cable Attachment



\* USB cable can be attached firmly to the converter by using a USB cable attachment.

#### ⚠ CAUTION

- The USB cable attachment cannot be used excluding a bundled cable.
- When the USB cable attachment is being used, do not remove or connect the USB cable on the converter side repeatedly. This may damage the USB cable attachment or yourself.

- 3** The hardware is automatically recognized by the OS and installation proceeds automatically.
- 4** You have now finished installing the hardware.

## 5. Device driver Initialization

Setting the device name is required to use the device driver. It is called the device driver initialization. The device name will be assigned automatically during hardware installation. Therefore, if you want to use it, you can skip the setting procedure described below.

How to check the device name and change the device name, refer to the help included in the development environment package you downloaded from the CONTEC website or “**Setting and Confirming Device Name**” in the online help made available on the CONTEC website.

### Online Help [Setting and Confirming Device Name]

<https://help.contec.com/link/drv/wdm/devicename/en/>

## 6. Operation Check

You must make sure that this product and device driver operate properly. By taking this step, you can be certain that this product has been set up appropriately.

### 1. Connection Method

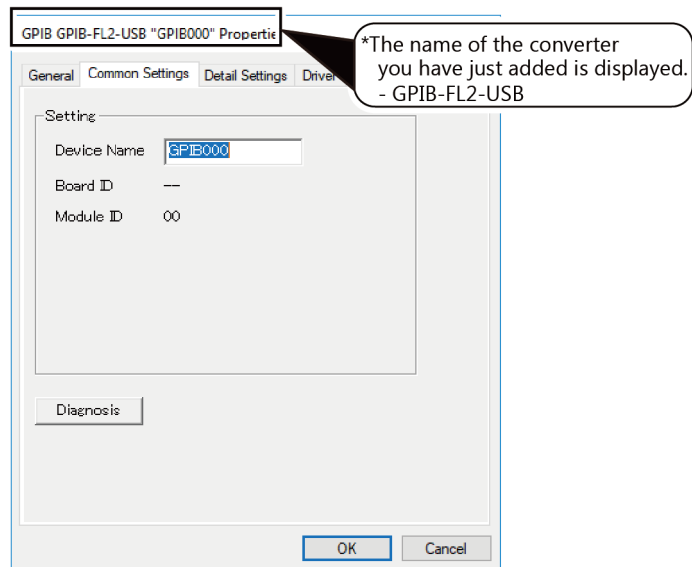
Connect the product to an external target device to test the communication and check the execution environment.

To connect an external target device, see **"Connecting to an External Device (page34)"**.

### 2. Using the Diagnosis Program

#### ◆ Starting the Diagnosis Program

Click the [Diagnosis] on the Properties page to start the diagnosis program.



## ◆ Checking Functions

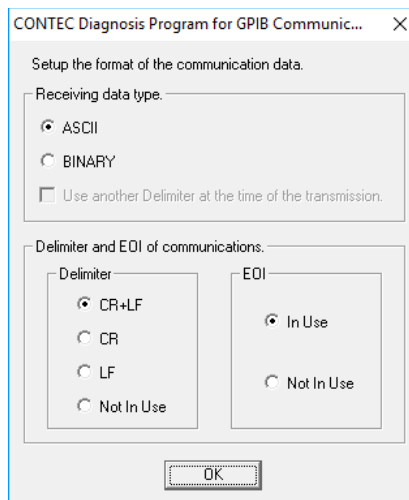
After the address and communication data format of the target device are set, the main dialog box of the Diagnosis Program appears.

You can check the current operation states of the board in the following boxes:

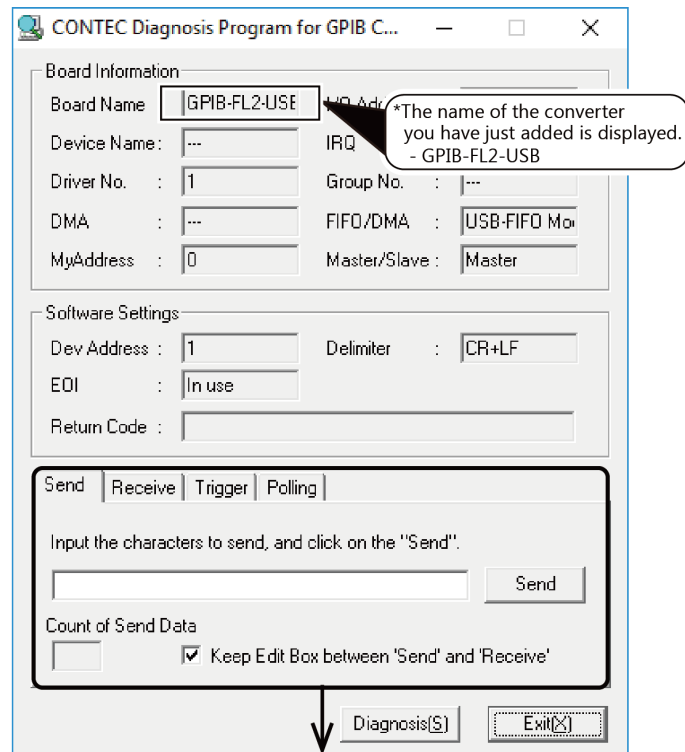
- 1 Specify the remote device address and click on the [OK] button.



- 2 Specify the communication data format and click on the [OK] button.



### 3 The main dialog box appears.



The following commands can be used to check operations.

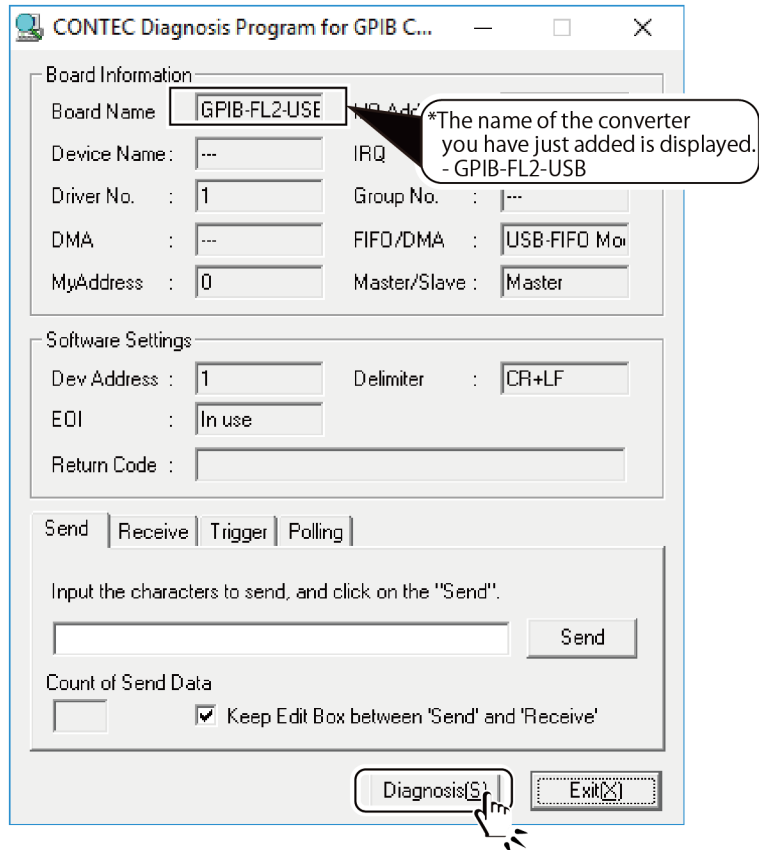
- "Send": Sends the typed character string to the remote device.
- "Receive": Receives data from the target device and displays it along with the number of data items.
- "Trigger": Sends a trigger command to the remote device.
- "Polling": Polls the remote device and displays the obtained status byte.

#### \*Note

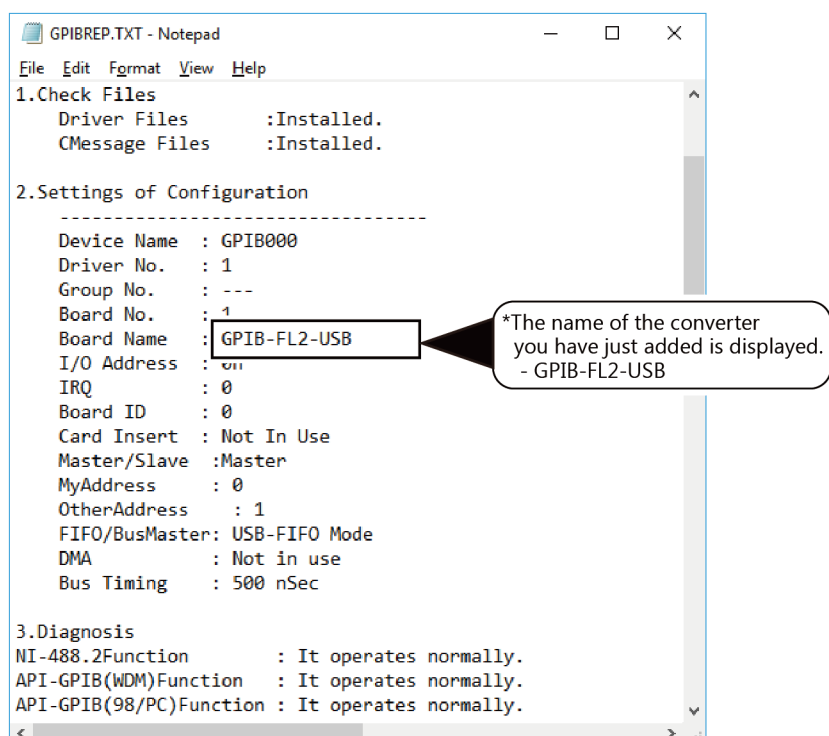
When communication has been completed successfully, "xxxxx completed normally" is displayed as the "return value".

## ◆ Diagnosis Report

- 1 Clicking on [Diagnosis] displays detailed data including converter settings and the diagnosis results while saving them in text format. The diagnosis program performs "driver file test", "GPIB communication test", "device setting information obtain" and so on.



- 2 A diagnosis report is displayed as shown below.



## 7.Setup Troubleshooting

### 1. Examples and Solution

---

#### ◆ The diagnostic program works properly but the application program does not.

The diagnostic program uses the API-GPIB(WDM) for USB functions. If the diagnostic program works properly, other applications should also work properly. If you have a problem, recheck your program taking note of the following points.

- Check the return values of the API functions.
- Refer to the source code for the sample programs.

#### ◆ If your problem cannot be resolved

Contact your retailer.

# Connecting to an External Device

This section describes how to connect the product to an external device with a cable.

# 1. Connecting to an External Device

## 1. Notes on GPIB cable connection

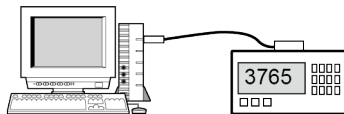
The GPIB has restrictions on the number of devices connected and the cable length according to the standard.

- The maximum number of interfaces (external devices) is 15, which can be connected to one system.
- The maximum total length of cables that can be used to interconnect a group of devices in one bus system is "2 m x (the number of devices)" or 20 m, whichever is shorter. (JIS C1901-1987)  
Note, however, the individual cables between devices must be within 4 m long.  
Some examples are given below.

### - System with a total of two devices

$2 \text{ m} \times (\text{Number of devices} = 2) < 20 \text{ m}$

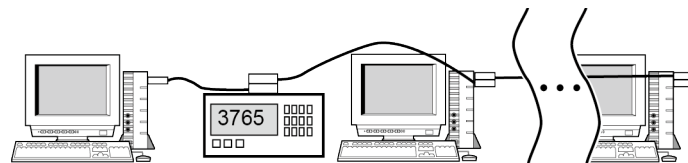
The maximum total length of cables for this system is therefore 4 m.



### - System with a total of three devices

$2 \text{ m} \times (\text{Number of devices} = 3) < 20 \text{ m}$

The maximum total length of cables for this system is therefore 6 m. The two cables used in the system must be  $[2 \text{ m} + 4 \text{ m}]$  or  $[2 \text{ m} + 2 \text{ m}]$  in length so that neither is longer than 4 m.

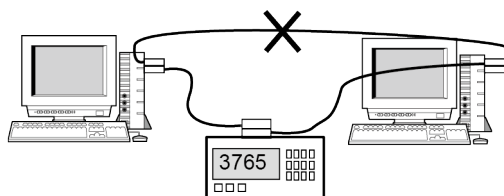


### - System with a total of fifteen devices

$2 \text{ m} \times (\text{Number of devices} = 15) > 20 \text{ m}$

The maximum total length of cables for this system is therefore 20 m.

- The cables in the system must not form a loop.



- Unplug the cable from any device which is left off for some reason such as a fault.
- When powering the measurement system, turn on the measuring instrument first and then on the PC.
- Neither unplug/plug the cable nor turn on/off the device during communication. Doing so stops the operation or causes an error, resulting in trouble.
- The talker and listener must be addressed to talk and to listen, respectively, by the controller before the talker can send messages to the listener.
- At least two-thirds of all the devices connected must be turned on.

# Software

This section describes GPIB communication driver for Windows developed by CONTEC and CONTEC Device Utility.

# 1.Application Development

The device driver listed below is provided for application development. The following sections describe each and the main programming languages that can be used.

## **API-GPIB(WDM) for USB**

It is the device driver for Windows with new function specification and supplies command in the form of standard Win32 API function (DLL). Settings of multiple GPIB communication converters can be done easily, therefore, install this software if you are developing a new application or are an existing user of the "GPIB Development Environment of API-USBP(WDM).

## **API-GPIB(98/PC)**

It is the device driver, and which supplies command in the form of standard Win32 API function (DLL). In addition, sample programs such as Visual Basic and Visual C++ are included, and you can verify the operation of hardware using Diagnostic programs. To replace our GPIB boat (F series) with this product, please install here.

## **API-GPLV(W32)**

API-GPLV(W32) is a driver created according to the National Instruments Corporation's GPIB function style. The driver is software to control the CONTEC GPIB board using a LabVIEW-based GPIB system or existing application program.

## 2. For using API-GPIB(WDM) for USB

When using the device driver API-GPIB(WDM) for USB for development, please refer to the online help and sample programs.

### 1. Accessing the Help File

It might be slightly different depending on the installed driver (development environment) version. From [Start] menu, click on [CONTEC API-USBP(WDM)]-[API-USBP(W32) Help] to display help information.

The information for application development, such as function reference is provided in [API-USBP(WDM) Help]. Detailed introduction to search method for help should be found from [How to navigate Help] in the help. Refer to the [Tutorial] in the help for basic usage.

### 2. Using Sample Programs

Sample programs are individually offered for the basic application.

The sample programs of the API-GPIB(WDM) for USB can be found in

\Program Files (x86)\CONTEC\API-USBP(WDM)\Gpib\Samples

or

\Program Files\CONTEC\API-USBP(WDM)\Gpib\Samples .

Use these sample programs as references for program development and operation check.

\* The sample program folder differs if you changed the install folder.

### 3. Running a Sample Program

From the Start Menu on the Windows taskbar, go to "Programs" – "CONTEC API-GPIB(WDM) for API-USBP(WDM)" and select the sample programs.

#### ◆ Sample Programs - Examples

- Master  
Executes a series of operations including transmit/receive, polling, and sending commands in master mode.
- Slave  
Executes a series of operations including transmit/receive, SRQ send, and response to parallel polling in slave mode.

## 4. Uninstalling the Device driver

---

Refer to the help file for detailed uninstalling procedure per OS.

### ◆ Uninstall of Device Driver

Refer to "**Uninstall of Device Driver (page44)**".

### ◆ Uninstall of Development Environment

Use [Control Panel] - [Programs and Features] to uninstall the development environment.  
Select [CONTEC API-USBP(WDM) GPIB Communication Develop] and then click [Uninstall].

## 3. For using API-GPIB(98/PC)

When using the device driver API-GPIB(98/PC) for development, please refer to the online help and sample programs.

### 1. Accessing the Help File

It might be slightly different depending on the installed driver (development environment) version. From [Start] menu, click on [CONTEC API-GPIB(98/PC) for API-PAC(W32)] –[API-GPIB HELP].

The information for application development, such as function reference is provided in [API-USBP(WDM) Help]. Refer to the [Tutorial] in the help for basic usage.

### 2. Using Sample Programs

Sample programs are individually offered for the basic application.

The sample programs of the API-GPIB(98/PC) can be found in

\Program Files (x86)\CONTEC\API-PAC(W32)\GPIB\Samples

or

\Program Files\CONTEC\API-PAC(W32)\GPIB\Samples.

Use these sample programs as references for program development and operation check.

\* The sample program folder differs if you changed the install folder.

### 3. Running a Sample Program

From the Start Menu on the Windows taskbar, go to "Programs" – "CONTEC API-GPIB(98/PC) for API-PAC(W32)" and select the sample programs.

#### ◆ Sample Programs - Examples

- Master  
Executes a series of operations including transmit/receive and polling in master mode.
- Slave  
Executes a series of operations including transmit/receive and SRQ send in slave mode.

## 4. Uninstalling the Device driver

---

Refer to the help file for detailed uninstalling procedure per OS.

### ◆ Uninstall of Device Driver

Refer to "**Uninstall of Device Driver (page44)**".

### ◆ Uninstall of Runtime Environment

Use [Control Panel] - [Programs and Features] to uninstall the development environment.  
Select [CONTEC API-GPIB(98/PC)NT VerX.XX (Runtime)] and then click [Uninstall].

### ◆ Uninstall of Development Environment

Use [Control Panel] - [Programs and Features] to uninstall the development environment.  
Select [CONTEC API-GPIB(98/PC)NT VerX.XX (Develop)] and then click [Uninstall].

## 4. For using API-GPLV(W32)

API-GPLV(W32) is a device driver created in NI's GPIB function style as the software for controlling CONTEC GPIB products.

When the device driver is installed, existing applications such as LabVIEW can operate CONTEC GPIB products. For details, refer to the help file.

### 1. Accessing the Help File

It might be slightly different depending on the installed driver (development environment) version. From [Start] menu, click on [CONTEC API-GPLV(W32) for API-PAC(W32)] – [API-GPLV HELP] to display help information.

The help file provides information such as "operation specifications", "additional information", and "troubleshooting".

### 2. Using Sample Programs

Sample programs are individually offered for the basic application.

The sample programs of the API-GPLV(W32) can be found in

\Program Files (x86)\CONTEC\API-PAC(W32)\Gplv\Samples

or

\Program Files\CONTEC\API-PAC(W32)\Gplv\Samples.

Use these sample programs as references for program development and operation check.

\* The sample program folder differs if you changed the install folder.

### 3. Running a Sample Program

From the Start Menu on the Windows taskbar, go to "Programs" – "CONTEC API-GPLV(W32) for API-PAC(W32)" and select the sample programs.

#### ◆ Sample Programs - Examples

- GpibTest  
Executes a series of operations including initialization, transmit/receive, and polling.

## 4. Uninstalling Device driver

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Refer to the help file for detailed uninstalling procedure per OS.

### ◆ Uninstall of Device Driver

Refer to "**Uninstall of Device Driver (page44)**".

### ◆ Uninstall of Runtime Environment

Use [Control Panel] - [Programs and Features] to uninstall the runtime environment.

Select [CONTEC API-GPLV(W32) VerX.XX (Runtime)] and then click [Uninstall].

### ◆ Uninstall of Development Environment

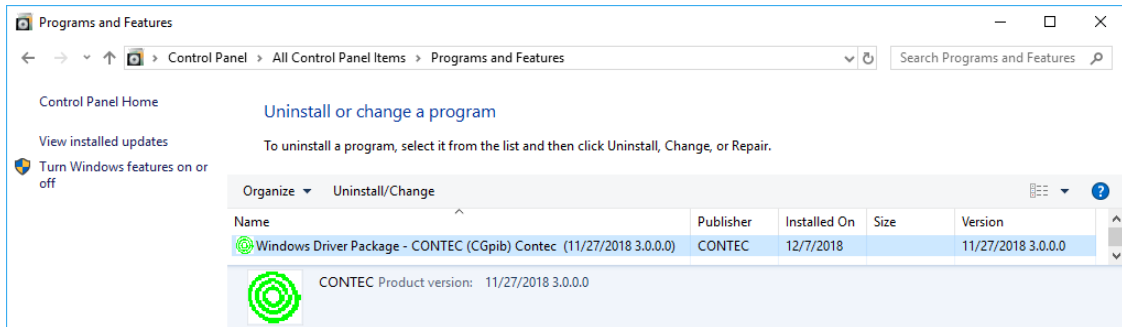
Use [Control Panel] - [Programs and Features] to uninstall the development environment.

Select [CONTEC API-GPLV(W32) VerX.XX (Develop)] and then click [Uninstall].

## 5. Uninstall of Device Driver

Refer to the help file for detailed uninstalling procedure per OS.

Use [Control Panel] - [Programs and Features] to uninstall the development environment.



Select [Windows driver package - CONTEC (CGpib) Contec] and then click [Uninstall/Change].

After the uninstallation described above has been completed, select [CONTEC API-GPIB(WDM) driver] and then click [Uninstall].

## 6.CONTEC Device Utility

CONTEC Device Utility is utility software to setup the CONTEC's measuring control device. Use this software to update firmware of the product.

### 1. Updating Firmware

Firmware is namely software which is embedded in the product. Up-to-date firmware (update file) will be supplied in the CONTEC website when function upgrades and so on.

The following presents how to update firmware.

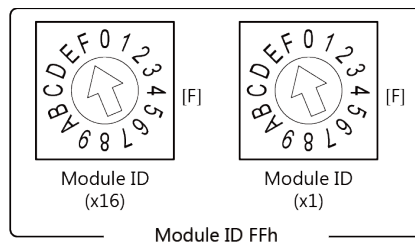
#### 1 Preparation for updating

Unplug and remove all of USB cables and interface connectors from the product once.

When using more than one converter, always connect each of them with a USB cable, and complete updating before starting to update the next product.

Set Module ID of this product to FFh.

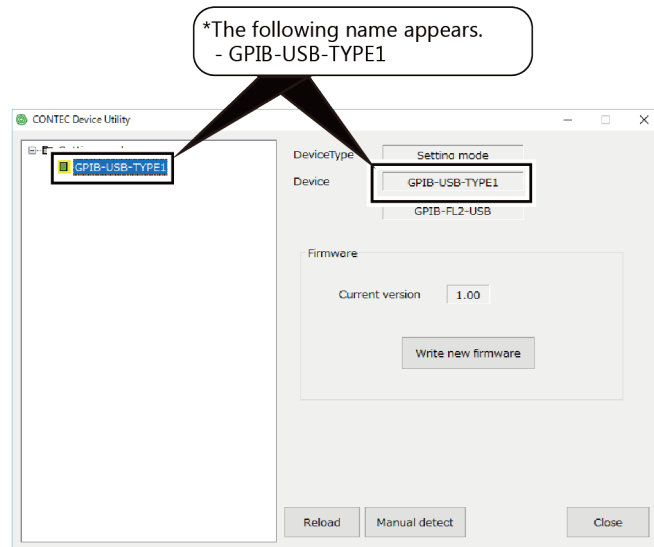
This setting is specifically used for firmware updating.



#### 2 Connect the product to USB port.

Connect the converter to USB port of the PC.

- 3** Start up the Utility.  
Start up CONTEC Device Utility.  
(The utility can be found in Utility folder of API-GPIB(WDM) for USB driver.)



- 4** Write firmware  
Select GPIB-USB-TYPE1 from the list, and perform updating firmware.
- 5** Remove the product.  
Remove the product from USB port of the PC.
- 6** Set Module ID back to the former ID.  
Set Module ID back to the ID that had been set before firmware was updated.

# Functions

This section describes the features achieved by combining hardware and device driver functions.

# 1.Basic GPIB Function

## ◆ Master/Slave function

The converter can be used as either the master (controller) or slave, depending on a setting in the property page.

When used as the master, the converter can send IFC (Interface Clear) at any timing and control the REN (Remote Enable) line.

## ◆ Communication function

The converter can send and receive data in accordance with the IEEE 488 Standard. You can add delimiters and EOI (End Of Identify) to outgoing data depending on the software settings.

## ◆ Serial poll/parallel poll/SRQ send functions

The following functions can be used depending on the master/slave configuration.

### Master

- Serial poll
- Parallel poll

### Slave

- Status byte setting
- SRQ (Service Request) transmission
- Response to parallel polling

## My address setting

The GPIB address (my address) of the converter can be set in Property page.

No setting is required on the converter.

## 2.Additional Functions

### ◆ Communication using FIFO memory

The product can use FIFO memory for communication. As the product controls this form of communication, it can be performed at high speed irrelevant to the PC's CPU speed.

Note, however, that the actual communication speed is set to the speed of the slowest device in compliance with the GPIB standard.

# Appendix

This section lists the specifications and the physical dimensions of the product.

# 1. Hardware Specification

## Function Specifications

| Item   |                               | Specification                                                 |
|--------|-------------------------------|---------------------------------------------------------------|
| GPIB   | The number of channels        | 1 channel<br>Conforms to IEEE-488.1, 488.2(GPIB)standards     |
|        | Transfer format               | 8-bit parallel, 3-wire handshake system                       |
|        | Transfer rate                 | 1.5Mbyte/sec                                                  |
|        | Data buffer size              | 2Kbyte send, 2Kbyte receive                                   |
|        | Signal logic                  | Negative logic L level : 0.8V or less, H level : 2.0V or more |
|        | Cable length between devices  | 4m or less *1                                                 |
|        | Total cable length            | 20m or less                                                   |
|        | Connectable number of devices | 15 devices (Max.)                                             |
| USB    | Bus specification             | USB Specification 2.0/1.1 standard                            |
|        | USB transfer rate             | 12Mbps (Full-speed), 480Mbps (High-speed) *2                  |
|        | Cable length                  | 1.8m                                                          |
|        | Power supply                  | Bus power                                                     |
| Common | Current consumption           | 5VDC 400mA (Max.)                                             |
|        | Physical dimensions (mm)      | 62(W) x 64(D) x 24(H) (No projection included)                |
|        | Weight                        | 100g (Not including the USB cable, attachment)                |

\*1 For details, see "**Notes on GPIB cable connection (page34)**".

\*2 Depends on the environment of the host PC (OS, USB host controller) being used.

## Installation Environment Requirements

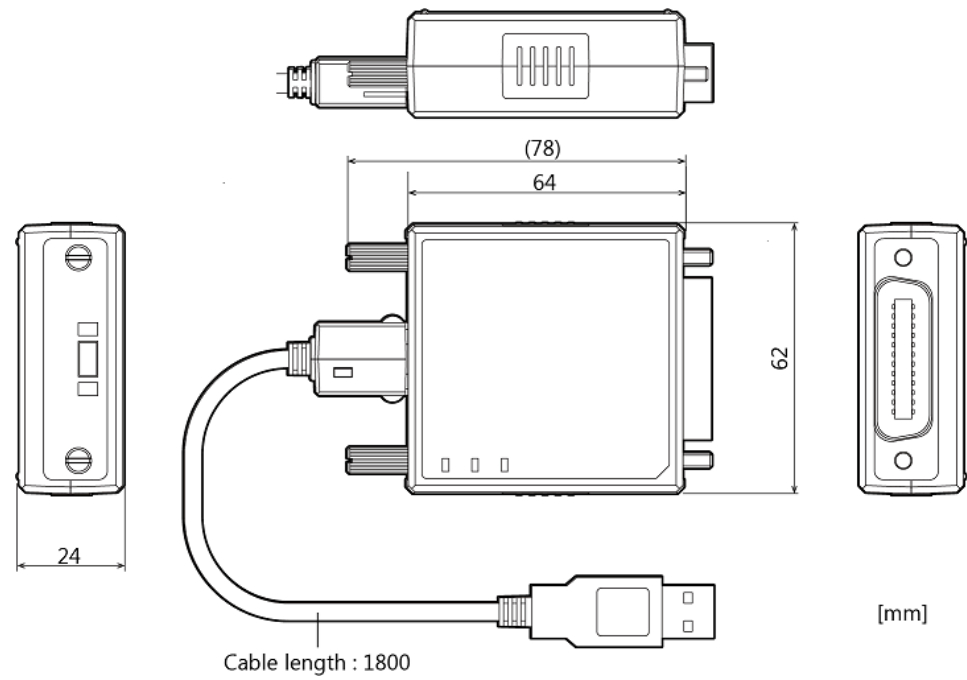
| Item                          |                               | Specification                                                                                                  |
|-------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Operating ambient temperature |                               | 0 - 50°C                                                                                                       |
| Operating ambient humidity    |                               | 10 - 90%RH (No condensation)                                                                                   |
| Floating dust particles       |                               | Not to be excessive                                                                                            |
| Corrosive gases               |                               | None                                                                                                           |
| Line-noise Resistance         | Line noise                    | AC Line/±2kV<br>Signal Line/±1kV (IEC61000-4-4 Level 3, EN61000-4-4 Level 3)                                   |
|                               | Static electricity resistance | Touch/±4kV (IEC61000-4-2 Level 2, EN61000-4-2 Level 2)<br>Air/±8kV (IEC61000-4-2 Level 3, EN61000-4-2 Level 3) |
| Vibration resistance          | Sweep resistance              | 40minutes each in X, Y, and Z directions (JIS C60068-2-6-compliant, IEC60068-2-6-compliant)                    |
| Shock resistance              |                               | 147m/s <sup>2</sup> (15G) half-sine shock for 11ms<br>JIS C60068-2-27-compliant, IEC60068-2-27-compliant)      |
| Standard                      |                               | VCCI Class A, FCC Class A,<br>CE Marking (EMC Directive Class A, RoHS Directive), UKCA                         |

## IEEE-488.1, IEEE-488.2(GPIB) standard complied

### The list of support functions

| Code | Function                                         |
|------|--------------------------------------------------|
| SH1  | Source handshake functions                       |
| AH1  | Acceptor handshake functions                     |
| T6   | Basic talker, serial polling, MLA talker release |
| L4   | Basic listener MTA listener release              |
| TE0  | No extended talker functions                     |
| LE0  | No extended listener functions                   |
| SR1  | Service request function                         |
| RL1  | Remote function                                  |
| DC1  | Device clear function                            |
| DT1  | Device trigger function                          |
| PP1  | Configuration by remote message                  |
| C1   | System controller function                       |
| C2   | IFC send                                         |
| C3   | REN send                                         |
| C4   | Response to SRQ                                  |
| C26  | Interface message send, parallel polling         |

## 2. Physical Dimensions



## 3. Differences between GPIB-FL2-USB and GP-IB(USB)FL

Applications developed with GP-IB(USB)FL can be used in GPIB-FL2-USB as the GPIB-FL2-USB is API-compatible with the GP-IB(USB)FL.

Some differences between the two are listed below, however, there are no changes in the electrical specifications.

|                     | <b>GP-IB(USB)FL</b>                            | <b>GPIB-FL2-USB</b>                            |
|---------------------|------------------------------------------------|------------------------------------------------|
| Data Transfer Mode  | INT, FIFO                                      | FIFO                                           |
| Current Consumption | 450mA                                          | 400mA                                          |
| Weight              | 110g (Not including the USB cable, attachment) | 100g (Not including the USB cable, attachment) |

# Customer Support and Inquiry

CONTEC provides the following support services for you to use CONTEC products more efficiently and comfortably.

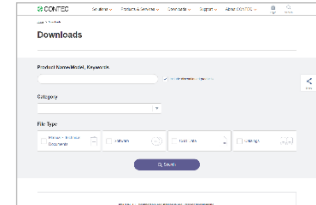
# 1.Services

CONTEC offers the useful information including product manuals that can be downloaded through the CONTEC website.

## Download

<https://www.contec.com/download/>

You can download updated device driver, firmware, and differential manuals in several languages. Membership registration (myCONTEC) is required to use the services.



## Revision History

| MONTH YEAR   | Summary of Changes                |
|--------------|-----------------------------------|
| January 2019 | The First Edition                 |
| April 2024   | Changed the layout of the manual. |
|              |                                   |
|              |                                   |
|              |                                   |

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<https://www.contec.com/>

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GPIO-FL2-USB Reference Manual

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